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Supplementary Materials for

Realization of three- and four-body interactions between momentum states in a cavity

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The PDF file includes:

Supplementary materials

1 Derivation of the effective three-body Hamiltonian

1.1 Second-order perturbation theory

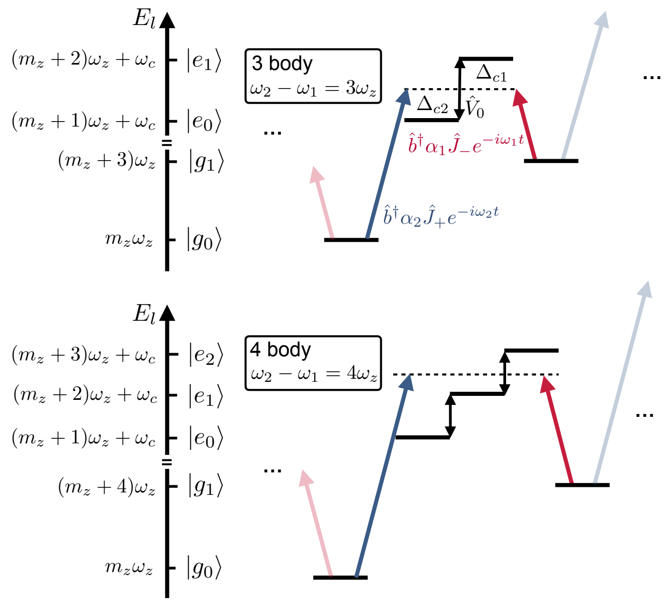

Figure S1: **Diagrammatic perturbation theory for multi-body interactions.** Upper panel: 3-body interaction mediated by a 6-photon process involving two intermediate states. The effective coupling process starts with the excitation of state $|g_0\rangle$ to $|e_0\rangle$ through the absorption of a blue pump photon. This is followed by the coupling between $|e_0\rangle$ and $|e_1\rangle$ via the operator $\hat{V}_0$, and finally, the system transitions to $|g_1\rangle$ through the emission of a red pump photon. Lower panel: 4-body interaction mediated by an 8-photon process involving three intermediate states.

In this section, we derive the 3-body interaction Hamiltonian presented in Eq. (1). After adiabatic elimination of the excited state [32,33,45], the system Hamiltonian is given by:

$$\begin{aligned}\hat{H}_S = \hat{H}_0 + \frac{g_0^2}{4\Delta_a}\hat{a}^\dagger\hat{a}\left(\hat{J}_+ + \hat{J}_-\right) \\ + \left(\epsilon_1 e^{-i\omega_1 t} + \epsilon_2 e^{-i\omega_2 t}\right)\hat{a}^\dagger + \text{h.c.},\end{aligned} \tag{S1}$$

here $\hat{H}_0 = \omega_c\hat{a}^\dagger\hat{a} + \omega_z\hat{J}_z$. The applied two dressing lasers have amplitudes $\epsilon_{1,2}$ at frequencies

$\omega_{1,2}$. We define $\mathcal{G}=\frac{g_0^2}{4\Delta_a}$ for simplicity and substitute the cavity field operator via:

$$\hat{a}=\hat{b}+\alpha_1 e^{-i\omega_1 t}+\alpha_2 e^{-i\omega_2 t} \tag{S2}$$

with the oscillating classical field built in the cavity given by $\alpha_{1,2}=\frac{\epsilon_{1,2}}{i\kappa/2+(\omega_{1,2}-\omega_c)}$ and $\hat{b}$ representing the cavity field quantum fluctuation. The atom-cavity coupling Hamiltonian is then given by $\hat{V}=\hat{V}_+ + \hat{V}_- + \hat{V}_0$, where

$$\begin{aligned}
&\hat{V}_+ = \mathcal{G}\hat{b}^\dagger\left(\alpha_1 e^{-i\omega_1 t}+\alpha_2 e^{-i\omega_2 t}\right)\left(\hat{J}_+ + \hat{J}_-\right)\\
&\hat{V}_- = \left(\hat{V}_+\right)^\dagger \quad \hat{V}_0 = \mathcal{G}\hat{b}^\dagger\hat{b}\left(\hat{J}_+ + \hat{J}_-\right)
\end{aligned} \tag{S3}$$

as the perturbation fields. Additionally, the cavity dissipation is described by $\hat{L}=\sqrt{\kappa}\hat{b}$.

We work with the basis states $|m_z, n_b\rangle$, where n_b represents the photon number in the cavity mode (with $\hat{b}^\dagger\hat{b}\,|m_z,n_b\rangle = n_b\,|m_z,n_b\rangle$). Here m_z is the eigenvalue of the collective pseudo-spin operator along the z-axis (with $\hat{J}_z\,|m_z,n_b\rangle = m_z\,|m_z,n_b\rangle$). Initially, we start in the state $|m_z, n_b=0\rangle$, corresponding to the ground state manifold with zero photons in mode $\hat{b}$. The operator $\hat{V}_+$ couples this state to the intermediate state manifold where $n_b=1$. We apply the general formula for second-order perturbation theory, as derived in Ref. [46], which involves multiple perturbation fields and non-perturbative coupling within the ground state manifold (with $\omega_z \sim |\omega_c-\omega_{1,2}|$), resulting in the following effective Hamiltonian and jump operator:

$$\begin{aligned}
\hat{H}_{\text{eff}} &= -\frac{1}{2}\left[\hat{V}_-\sum_{f,l}\left(\hat{H}_{\text{NH}}^{(f,l)}\right)^{-1}\hat{V}_+^{(f,l)}(t)+\text{h.c.}\right]+\hat{H}_g\\
\hat{L}_{\text{eff}} &= \hat{L}\sum_{f,l}\left(\hat{H}_{\text{NH}}^{(f,l)}\right)^{-1}\hat{V}_+^{(f,l)}(t)\,.
\end{aligned} \tag{S4}$$

Here, the index f represents the perturbation fields with frequency ω_f, l denotes the ground state with energy E_l, and $\hat{H}_{\text{NH}}^{(f,l)} = \hat{H}_{\text{NH}} - E_l - \omega_f$ with $\hat{H}_{\text{NH}} = \hat{H}_e - \frac{1}{2}\hat{L}^\dagger\hat{L}$ is the non-Hermitian Hamiltonian for the intermediate state manifold.

We consider a four-level model where $|g_0\rangle \equiv |m_z,0\rangle$, $|e_0\rangle \equiv |m_z+1,1\rangle$, $|e_1\rangle \equiv |m_z+2,1\rangle$ and $|g_1\rangle \equiv |m_z+3,0\rangle$ with the ground states $|g_{0,1}\rangle$ and intermediate state $|e_{0,1}\rangle$ shown in

Fig. S1, therefore the non-Hermition Hamiltonian is given by:

$$\begin{aligned}\hat{H}_{\mathrm{NH}} = &\left[(m_z+1)\,\omega_z + \omega_c - i\frac{\kappa}{2}\right] |e_0\rangle\langle e_0| + \left[(m_z+2)\,\omega_z + \omega_c - i\frac{\kappa}{2}\right] |e_1\rangle\langle e_1| \\ &+ \mathcal{G} f_{m_z+1} \left(|e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|\right)\end{aligned} \tag{S5}$$

where $f_{m_z} = \sqrt{\frac{N}{2}(\frac{N}{2}+1) - m_z(m_z+1)}$ representing the coupling factor between Dicke states $|m_z\rangle$ and $|m_z+1\rangle$. Also, we have the perturbation terms:

$$\hat{V}_+ = \mathcal{G}\left(\alpha_1 e^{-i\omega_1 t} + \alpha_2 e^{-i\omega_2 t}\right)\left(f_{m_z}|e_0\rangle\langle g_0| + f_{m_z+2}|e_1\rangle\langle g_1|\right). \tag{S6}$$

As a result, there exist four $\hat{H}_{\mathrm{NH}}^{(f,l)}$ for various frequencies ω_f and ground levels l:

$$\begin{aligned}\hat{H}_{\mathrm{NH}}^{(f=1,l=g_0)} &= \left(-\Delta_{c1} + 2\omega_z - i\frac{\kappa}{2}\right)|e_0\rangle\langle e_0| + \left(-\Delta_{c1} + 3\omega_z - i\frac{\kappa}{2}\right)|e_1\rangle\langle e_1| \\ &\quad + \mathcal{G} f_{m_z+1}\left(|e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|\right) \\ \hat{H}_{\mathrm{NH}}^{(f=2,l=g_0)} &= \left(-\Delta_{c2} - i\frac{\kappa}{2}\right)|e_0\rangle\langle e_0| + \left(-\Delta_{c2} + \omega_z - i\frac{\kappa}{2}\right)|e_1\rangle\langle e_1| \\ &\quad + \mathcal{G} f_{m_z+1}\left(|e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|\right) \\ \hat{H}_{\mathrm{NH}}^{(f=1,l=g_1)} &= \left(-\Delta_{c1} - \omega_z - i\frac{\kappa}{2}\right)|e_0\rangle\langle e_0| + \left(-\Delta_{c1} - i\frac{\kappa}{2}\right)|e_1\rangle\langle e_1| \\ &\quad + \mathcal{G} f_{m_z+1}\left(|e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|\right) \\ \hat{H}_{\mathrm{NH}}^{(f=2,l=g_1)} &= \left(-\Delta_{c2} - 3\omega_z - i\frac{\kappa}{2}\right)|e_0\rangle\langle e_0| + \left(-\Delta_{c2} - 2\omega_z - i\frac{\kappa}{2}\right)|e_1\rangle\langle e_1| \\ &\quad + \mathcal{G} f_{m_z+1}\left(|e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|\right).\end{aligned} \tag{S7}$$

Each term has a clear physical meaning, for example, $\hat{H}_{\mathrm{NH}}^{(f=1,l=g_0)}$ corresponds to couple from $|g_0\rangle$ to the intermediate states via the perturbation field at frequency ω_1. Notice that $\Delta_{c2} - \Delta_{c1} = \omega_z$ thus $f = 2, l = e_0$ and $f = 1, l = e_1$ share the same non-Hermitian Hamiltonian for the intermediate states.

Above non-Hermitian Hamiltonians take the form of $\begin{pmatrix} \tilde{E}_0 & \mathcal{G} f_{m_z+1} \\ \mathcal{G} f_{m_z+1} & \tilde{E}_1 \end{pmatrix}$ in the $|e_{0,1}\rangle$ basis, as a result, the inversions can be evaluated as $\frac{1}{\tilde{E}_0\tilde{E}_1 - (\mathcal{G} f_{m_z+1})^2}\begin{pmatrix} \tilde{E}_1 & -\mathcal{G} f_{m_z+1} \\ -\mathcal{G} f_{m_z+1} & \tilde{E}_0 \end{pmatrix}$. Then we calculate the contribution to the effective Hamiltonian from individual perturbation fields and energy levels as $\hat{H}_{\text{eff}}^{(f,l)} = -\frac{1}{2}\hat{V}_{-}\left(\hat{H}_{\text{NH}}^{(f,l)}\right)^{-1}\hat{V}_{+}^{(f,l)} + \text{h.c.}$:

$$\begin{aligned}
\hat{H}_{\text{eff}}^{(f=1,l=g_0)} &= \frac{\mathcal{G}^2\left|\alpha_1\right|^2(\Delta_{c1}-2\omega_z)}{(\Delta_{c1}-2\omega_z)^2+(\kappa/2)^2} f_{m_z}^2 \left|g_0\right\rangle\left\langle g_0\right| \\
\hat{H}_{\text{eff}}^{(f=2,l=g_0)} &= \frac{\mathcal{G}^3\alpha_1^*\alpha_2 f_{m_z}f_{m_z+1}f_{m_z+2}}{2(\Delta_{c1}+i\kappa/2)(\Delta_{c2}+i\kappa/2)}\left|g_1\right\rangle\left\langle g_0\right| e^{-3i\omega_z t} + \text{h.c.} + \frac{\mathcal{G}^2\left|\alpha_2\right|^2\Delta_{c2}}{\Delta_{c2}^2+(\kappa/2)^2} f_{m_z}^2\left|g_0\right\rangle\left\langle g_0\right| \\
\hat{H}_{\text{eff}}^{(f=1,l=g_1)} &= \frac{\mathcal{G}^3\alpha_2\alpha_1^* f_{m_z}f_{m_z+1}f_{m_z+2}}{2(\Delta_{c1}+i\kappa/2)(\Delta_{c2}+i\kappa/2)}\left|g_0\right\rangle\left\langle g_1\right| e^{3i\omega_z t} + \text{h.c.} + \frac{\mathcal{G}^2\left|\alpha_1\right|^2\Delta_{c1}}{\Delta_{c1}^2+(\kappa/2)^2} f_{m_z+2}^2\left|g_1\right\rangle\left\langle g_1\right| \\
\hat{H}_{\text{eff}}^{(f=2,l=g_1)} &= \frac{\mathcal{G}^2\left|\alpha_2\right|^2(\Delta_{c2}+2\omega_z)}{(\Delta_{c2}+2\omega_z)^2+(\kappa/2)^2} f_{m_z+2}^2\left|g_1\right\rangle\left\langle g_1\right|.
\end{aligned} \tag{S8}$$

Notice that we ignore the coupling terms $\left|g_1\right\rangle\left\langle g_0\right|$ in $\hat{H}_{\text{eff}}^{(f=1,l=g_0)}$ and $\hat{H}_{\text{eff}}^{(f=2,l=g_1)}$, as they oscillate with $e^{3i\omega_z t}$ and are therefore negligible in the atomic rotating frame descried by the Hamiltonian $\omega_z\hat{J}_z$. Also we assume $\mathcal{G}\sqrt{N} \ll \Delta_{c1,2}$ in the experiment, allowing us to neglect the $(\mathcal{G}f_{m_z+1})^2$ terms in the denominators (they contribute to even high-order terms like $\hat{J}_-^2\hat{J}_+^2$).

A similar treatment can be applied to each Dicke state $\left|m_z\right\rangle$, and summing over m_z yields the effective Hamiltonian:

$$\begin{aligned}
\hat{H}_{\text{eff}} \approx\ & \mathcal{G}^3\alpha_1^*\alpha_2 \text{Re}\frac{1}{(\Delta_{c1}+i\kappa/2)(\Delta_{c2}+i\kappa/2)}\hat{J}_+^3 e^{-3i\omega_z t} + \text{h.c.} \\
& + \mathcal{G}^2\left(\frac{\left|\alpha_1\right|^2\Delta_{c1}}{\Delta_{c1}^2+(\kappa/2)^2} + \frac{\left|\alpha_2\right|^2(\Delta_{c2}+2\omega_z)}{(\Delta_{c2}+2\omega_z)^2+(\kappa/2)^2}\right)\hat{J}_+\hat{J}_- \\
& + \mathcal{G}^2\left(\frac{\left|\alpha_1\right|^2(\Delta_{c1}-2\omega_z)}{(\Delta_{c1}-2\omega_z)^2+(\kappa/2)^2} + \frac{\left|\alpha_2\right|^2\Delta_{c2}}{\Delta_{c2}^2+(\kappa/2)^2}\right)\hat{J}_-\hat{J}_+ \\
& + \omega_z\hat{J}_z
\end{aligned} \tag{S9}$$

The first line describes the 3-body interaction term. In the frame co-rotating with the atomic coherence, this term becomes time-independent. The second and third lines represent the exchange interaction, which vanishes under the experimental conditions $\Delta_{c1} = -\Delta_{c2}$ and $\left|\alpha_1\right| = \left|\alpha_2\right|$.

The effective jump operators are given by:

$$\begin{aligned}\hat{L}_{\text{eff}} = & \mathcal{G}\sqrt{\kappa}\alpha_1\left(\frac{\hat{J}_+e^{-i\omega_1 t}}{-\Delta_{c1}+2\omega_z-i\frac{\kappa}{2}}+\frac{\hat{J}_-e^{-i\omega_1 t}}{-\Delta_{c1}-i\frac{\kappa}{2}}\right)\\ &+\mathcal{G}\sqrt{\kappa}\alpha_2\left(\frac{\hat{J}_+e^{-i\omega_2 t}}{-\Delta_{c2}-i\frac{\kappa}{2}}+\frac{\hat{J}_-e^{-i\omega_2 t}}{-\Delta_{c2}-2\omega_z-i\frac{\kappa}{2}}\right)\\ &+\frac{\mathcal{G}^2}{\left(\Delta_{c1}+i\kappa/2\right)\left(\Delta_{c2}+i\kappa/2\right)}\sqrt{\kappa}\left(\alpha_1 e^{-i\omega_1 t}\hat{J}_-^2+\alpha_2 e^{-i\omega_2 t}\hat{J}_+^2\right).\end{aligned} \tag{S10}$$

The first and second lines represent the superradiance process, where the atom absorbs a pump photon and subsequently the cavity photon leaks out of the cavity [32]. The third line arises from a higher-order process, where the atom first absorbs a pump photon, then evolves under the operator $\hat{V}_0$, and finally, the cavity photon leaks out of the cavity. This process is typically of order $\mathcal{G}^2/\Delta_{c1,2}^2$, which is much weaker than the superradiance and can be neglected under current experimental conditions.

Since in a frame rotating with the atomic coherence, each term oscillates at distinct frequency, we can treat them as independent dissipation channels without interference. Under the condition $\Delta_{c1} = -\Delta_{c2}$ and $|\alpha_1| = |\alpha_2|$, the superradiance processes become balanced, yielding $\hat{L}_+ = \sqrt{\Gamma}\hat{J}_+$ and $\hat{L}_- = \sqrt{\Gamma}\hat{J}_-$ where $\Gamma \approx \mathcal{G}^2\kappa|\alpha_1|^2/(\Delta_{c1}^2 + (\kappa/2)^2)$. We can then compute the Heisenberg equations of motion of some spin observables:

$$\begin{aligned}\frac{d}{dt}\left\langle\hat{J}_x\right\rangle &= \cdots - \Gamma\left\langle\hat{J}_x\right\rangle\\ \frac{d}{dt}\left\langle\hat{J}_y\right\rangle &= \cdots - \Gamma\left\langle\hat{J}_y\right\rangle\\ \frac{d}{dt}\left\langle\hat{J}_z\right\rangle &= \cdots - 2\Gamma\left\langle\hat{J}_z\right\rangle\\ \frac{d}{dt}\left\langle\hat{J}_x^2\right\rangle &= \cdots + 2\Gamma\left(\left\langle\hat{J}_z^2\right\rangle - \left\langle\hat{J}_x^2\right\rangle\right)\\ \frac{d}{dt}\left\langle\hat{J}_y^2\right\rangle &= \cdots + 2\Gamma\left(\left\langle\hat{J}_z^2\right\rangle - \left\langle\hat{J}_y^2\right\rangle\right)\\ \frac{d}{dt}\left\langle\hat{J}_z^2\right\rangle &= \cdots + 2\Gamma\left(\left\langle\hat{J}^2\right\rangle - 3\left\langle\hat{J}_z^2\right\rangle\right).\end{aligned} \tag{S11}$$

Here, $\cdots$ represents the contributions from the unitary dynamics. It is worth noting that due to the balance in the superradiance processes, the collective enhancement for superradiance disappears, resulting in a much slower time scale (by $O(N)$) compared to the collective dynamics.

A similar approach can be applied to derive the 4-body interaction shown in Fig. S1, where we must consider three intermediate states coupled via the operator $\hat{V}_0$, and the non-Hermitian Hamiltonian $\hat{H}_{\rm NH}$ becomes a 3×3 matrix. Higher-order processes, such as terms like $\hat{J}_-^2 \hat{J}_+^2$, can occur in the system. However, only the term $\hat{J}_+^4$ accounts for the four-cycle oscillation observed in Fig. 4. On the other hand, the non-symmetrical pump scheme in this case leads to unbalanced superradiance, which becomes significant, as indicated by the observed shift of the signal away from $J_z = 0$.

1.2 Average Hamiltonian theory

Even though second-order perturbation has been widely used in the past for the derivation of effective Hamiltonians in the cold gases and trapped ions communities, we also provide an alternative derivation using a third-order expansion, a procedure which has been employed recently to the derivation of more complex effective Hamiltonians emerging from the applications of multiple tones [22,23] In this section, we present an alternative derivation based on well-known time-averaging Hamiltonian theory [47] extending to higher-order, which yields consistent results when setting $\kappa = 0$. We start with a general time-dependent Hamiltonian $\hat{H}_I = \sum_i \hat{h}_i^\dagger e^{i f_i t} + h.c.$ ($f_i > 0$) and unitary time evolution operator $\hat{U}(t,0)$ satisfying $i\hbar \frac{\partial}{\partial t}\hat{U}(t,0) = \hat{H}_I(t)\hat{U}(t,0)$, the time-averaged Hamiltonian is expressed as [47]:

$$H_{\rm eff} = \frac{1}{2}\{\mathcal{H}_{\rm eff}(t) + \mathcal{H}_{\rm eff}^\dagger(t)\} \quad \mathcal{H}_{\rm eff}(t) = \{\overline{\hat{H}_I(t)\hat{U}(t,0)}\}\{\overline{\hat{U}(t,0)}\}^{-1}. \tag{S12}$$

Here $\overline{\cdots}$ denotes a time average, and the fast-rotating terms are discarded. The time-order expansion for $\hat{U}(t,0)$ is given by:

$$\begin{aligned}\hat{U}(t,0) &= \hat{T}\exp\left[-i\int_0^t \hat{H}_I(t')dt'\right] \\ &= I + \frac{1}{i\hbar}\int_0^t \hat{H}_I(t_1)dt_1 - \frac{1}{\hbar^2}\int_0^t dt_1 \int_0^{t_1} dt_2 \hat{H}_I(t_1)\hat{H}_I(t_2) + \cdots\end{aligned} \tag{S13}$$

and we can define the first and second-order propergators,

$$\hat{U}_1(t) = \frac{1}{i\hbar}\int_0^t \hat{H}_I \quad \hat{U}_2(t) = -\frac{1}{\hbar^2}\int_0^t dt_1 \int_0^{t_1} dt_2 \hat{H}_I(t_1)\hat{H}_I(t_2) \tag{S14}$$

Moreover, the properties $\hat{H}_I^\dagger(t) = \hat{H}_I(t)$ and $\hat{U}_1^\dagger(t) = -\hat{U}_1(t)$ hold, and under the time average $\overline{\hat{H}_I(t)} = 0$, $\overline{\hat{U}_1(t)} = \sum_i \frac{\hat{h}_i^\dagger - \hat{h}_i}{f_i}$.

Retaining terms up to third order in $H_I(t)$, we find

$$\begin{aligned}\mathcal{H}_{\text{eff}} &\approx \overline{\hat{H}_I(t)\{I + \hat{U}_1(t) + \hat{U}_2(t)\}} \times \overline{\{I + \hat{U}_1(t) + \hat{U}_2(t)\}}^{-1} \\ &\approx \overline{\hat{H}_I(t)\hat{U}_1(t)} + \overline{\hat{H}_I(t)\hat{U}_2(t)} - \overline{\hat{H}_I(t)\hat{U}_1(t)} \times \overline{\hat{U}_1(t)}.\end{aligned} \tag{S15}$$

Thus, the effective Hamiltonian is expressed as

$$\begin{aligned}H_{\text{eff}} = \frac{1}{2}(\mathcal{H}_{\text{eff}} + \mathcal{H}_{\text{eff}}^\dagger) &= \frac{1}{2}\{\overline{[\hat{H}_I(t), \hat{U}_1(t)]} + \overline{\hat{H}_I(t)\hat{U}_2(t)} + \overline{\hat{U}_2^\dagger(t)\hat{H}_I(t)} \\ &- \overline{\hat{H}_I(t)\hat{U}_1(t)} \times \overline{\hat{U}_1(t)} - \overline{\hat{U}_1(t)} \times \overline{\hat{U}_1(t)\hat{H}_I(t)}\}.\end{aligned} \tag{S16}$$

The first term corresponds to the second-order contribution ($\mathcal{G}^2$) while the remaining terms represent the third-order contribution ($\mathcal{G}^3$) in the effective Hamiltonian.

Back to our case, in the interaction picture of $\hat{H}_0 = \omega_c \hat{b}^\dagger\hat{b} + \omega_z \hat{J}_z$, the atom-cavity coupling Hamiltonian is expressed as:

$$\begin{aligned}\hat{H}_I/\hbar &= \mathcal{G}\hat{a}^\dagger\hat{a}(\hat{J}_+ e^{i\omega_z t} + \hat{J}_- e^{-i\omega_z t}) \\ &= \mathcal{G}(\hat{b}^\dagger e^{i\omega_c t} + \alpha_1^* e^{i\omega_1 t} + \alpha_2^* e^{i\omega_2 t})(\hat{b}e^{-i\omega_c t} + \alpha_1 e^{-i\omega_1 t} + \alpha_2 e^{-i\omega_2})(\hat{J}_+ e^{i\omega_z t} + \hat{J}_- e^{-i\omega_z t}).\end{aligned} \tag{S17}$$

Then we can decomposed $\hat{H}_I(t)$ via:

$$
\begin{aligned}
&\hat{h}_0^\dagger = \mathcal{G}(\hat{b}^\dagger\hat{b} + |\alpha_1|^2 + |\alpha_2|^2)\hat{J}_+ \quad f_0 = \omega_z \\
&\hat{h}_1^\dagger = \mathcal{G}\alpha_1\hat{b}^\dagger\hat{J}_- \quad f_1 = -\Delta_{c1} \equiv \omega_c - \omega_1 - \omega_z \\
&\hat{h}_2^\dagger = \mathcal{G}\alpha_2^*\hat{b}\hat{J}_- \quad f_2 = \Delta_{c2} \equiv \omega_2 - \omega_c - \omega_z \\
&\hat{h}_3^\dagger = \mathcal{G}\alpha_2^*\alpha_1\hat{J}_+ \quad f_3 = 4\omega_z \\
&\hat{h}_4^\dagger = \mathcal{G}\alpha_2^*\alpha_1\hat{J}_- \quad f_4 = 2\omega_z \\
&\hat{h}_5^\dagger = \mathcal{G}\alpha_1\hat{b}^\dagger\hat{J}_+ \quad f_5 = -\Delta_{c1} + 2\omega_z = \omega_c - \omega_1 + \omega_z \\
&\hat{h}_6^\dagger = \mathcal{G}\alpha_2^*\hat{b}\hat{J}_+ \quad f_6 = \Delta_{c2} + 2\omega_z = \omega_2 - \omega_c + \omega_z.
\end{aligned}
\tag{S18}
$$

The physical interpretation of each term is straightforward; for instance, $\hat{h}_{1,5}$ represents the blue and red modulation sideband for the first pump. In our pump scheme, we set $\omega_2 - \omega_1 = 3\omega_z$, thus $f_2 + f_1 = \omega_z$ and $|f_2 - f_1| \equiv \delta \ll \omega_z, \Delta_{c1,2}$. All other frequency components vanish under time averaging, except for $f_2 - f_1$.

Next, we apply the formula introduced in Eq. (S16). For the second-order terms, we obtain

$$
\begin{aligned}
\frac{1}{2}\overline{[\hat{H}_I(t), \hat{U}_1(t)]} &= \frac{1}{2}\overline{[\sum_i \hat{h}_i^\dagger e^{if_it} + \hat{h}_i e^{-if_it}, -\sum_i \hat{h}_i^\dagger \frac{e^{if_it}}{f_i} + \hat{h}_i \frac{e^{-if_it}}{f_i}]} \\
&= \sum_i \frac{[\hat{h}_i^\dagger, \hat{h}_i]}{f_i} + \frac{[\hat{h}_1^\dagger, \hat{h}_2]e^{i(f_1-f_2)t}}{\bar{f}_{12}} + \frac{[\hat{h}_2^\dagger, \hat{h}_1]e^{i(f_2-f_1)t}}{\bar{f}_{12}} \\
&\approx \frac{\mathcal{G}^2|\alpha_2|^2}{\Delta_{c2}}\hat{J}_-\hat{J}_+ + \frac{\mathcal{G}^2|\alpha_2|^2}{\Delta_{c2}+2\omega_z}\hat{J}_+\hat{J}_- + \frac{\mathcal{G}^2|\alpha_1|^2}{\Delta_{c1}}\hat{J}_+\hat{J}_- + \frac{\mathcal{G}^2|\alpha_1|^2}{\Delta_{c1}-2\omega_z}\hat{J}_-\hat{J}_+,
\end{aligned}
\tag{S19}
$$

where $\frac{2}{\bar{f}_{ij}} = \frac{1}{f_i} + \frac{1}{f_j}$ for $i \neq j$. These terms are dominated by the exchange interaction, as anticipated [32,43]. Such exchange interactions can be precisely canceled by setting $\Delta_{c1} = -\Delta_{c2}$ and $|\alpha_1| = |\alpha_2|$.

For the third-order term after some math, we find

$$
\frac{1}{2}\{\overline{\hat{H}_I(t)\hat{U}_1(t)} \times \overline{\hat{U}_1(t)} + \overline{\hat{U}_1(t)} \times \overline{\hat{U}_1(t)\hat{H}_I(t)}\} = \frac{1}{2}[\sum_i \frac{[\hat{h}_i^\dagger, \hat{h}_i]}{f_i}, \sum_j \frac{\hat{h}_j^\dagger - \hat{h}_j}{\omega_j}],
\tag{S20}
$$

and

$$\frac{1}{2}\{\overline{\hat{H}_I(t)\hat{U}_2(t)}+\overline{\hat{U}_2^\dagger(t)\hat{H}_I(t)}\}=\frac{1}{2}[\sum_i\frac{[\hat{h}_i^\dagger,\hat{h}_i]}{f_i},\sum_j\frac{\hat{h}_j^\dagger-\hat{h}_j}{\omega_j}]$$

$$+\frac{f_1[[\hat{h}_0,\hat{h}_1^\dagger],\hat{h}_2^\dagger]+f_1[[\hat{h}_0^\dagger,\hat{h}_1],\hat{h}_2]+f_2[[\hat{h}_0,\hat{h}_2^\dagger],\hat{h}_1^\dagger]+f_2[[\hat{h}_0^\dagger,\hat{h}_2],\hat{h}_1]}{f_1f_2(f_1+f_2)} \tag{S21}$$

$$=\frac{1}{2}[\sum_i\frac{[\hat{h}_i^\dagger,\hat{h}_i]}{f_i},\sum_j\frac{\hat{h}_j^\dagger-\hat{h}_j}{\omega_j}]+\frac{U^3}{\Delta_{c1}\Delta_{c2}}(\alpha_2^*\alpha_1\hat{J}_+^3+\alpha_2^*\alpha_1\hat{J}_-^3)$$

As a result, the effective Hamiltonian is given by:

$$H_{\text{eff}}=\frac{\mathcal{G}^3}{\Delta_{c1}\Delta_{c2}}(\alpha_2\alpha_1^*\hat{J}_+^3+\alpha_1\alpha_2^*\hat{J}_-^3). \tag{S22}$$

2 Generation of Bragg and dressing laser tones

We use a single laser for driving Bragg rotation and inducing the interaction to meet the phase-matching requirement [32]. Driving the Bragg rotations between $|p_0-\hbar k\rangle$ and $|p_0+\hbar k\rangle$ requires two different optical tones on the laser separated by ω_z. We realize this by first red shift the laser with one AOM and then blue shift it back with another AOM driven with two radio frequency (RF) tones at $\omega_{RF1}=2\pi\times 75$ MHz and $\omega_{RF2}=2\pi\times 75\ \text{MHz}-\omega_z$. Because the atoms accelerate under gravity, we linearly chirp the frequency separation $\omega_z/2\pi$ between the two sidebands at a rate $\delta_r=2\pi\times 25.11$ kHz/ms to compensate the changing Doppler shift. We frequency stabilize both Bragg laser tones roughly 3 MHz from the cavity resonance, such that both tones non-resonantly enter the cavity and drive the Bragg rotations. Since the detuning $3\ \text{MHz}\gg\kappa/2\pi$, we suppress any frequency noise to amplitude or phase noise conversion that would degrade the fidelity of the Bragg rotations. For all of the Bragg pulses applied in the interferometer sequence, the Rabi frequency is 8 kHz, giving a π-pulse duration of $63.$ μs.

For driving the 3-body interactions, we need two dressing laser tones separated by $3\omega_z$ while being phase-coherent with respect to the Bragg coupling. To achieve that, we first triple the frequency of ω_{RF2} and double the frequency of ω_{RF1}. We then mix the multiplied signals together to generate a third tone at frequency $\omega_{RF3} = 3\omega_{RF2} - 2\omega_{RF1} = 2\pi \times 75\ \mathrm{MHz} - 3\omega_z$. We combine this new RF tone with $\omega_{RF1} = 2\pi \times 75$ MHz to drive an AOM (acousto-optic modulator) to create two optical laser tones separated by $3\omega_z$. Since this new RF frequency tone is generated from the same frequency sources used to drive the Bragg transitions ω_{RF1} and ω_{RF2}, the interaction phase is thus phase stabilized to the Bragg coupling.

The driving laser tones for 4-body interaction can be engineered in a similar way. In this case, we quadrupole the frequency of ω_{RF2}, triple the frequency of ω_{RF1} and mix them together to get $\omega_{RF3} = 4\omega_{RF2} - 3\omega_{RF1} = 2\pi \times 75\ \mathrm{MHz} - 4\omega_z$. We then combine this RF tone with ω_{RF1} and drive an AOM to create two optical laser tones separated by $4\omega_z$ for driving the 4-body interactions with a well defined phase relationship to the Bragg coupling.

In every run of the experiment, the second RF tones ω_{RF2} starts chirping with respect to the first RF tone ω_{RF2} at $t = 0$ when the atoms start falling. For the first Bragg pulse that creates the superposition of momentum states, the relative phase between the two RF tones defines the phase for the Bragg coupling $\phi_B = \phi_{RF2} - \phi_{RF1} = \omega_z t + \pi\delta_r t^2$. For driving the interaction, the two dressing laser tones are created by the RF tones ω_{RF1} and ω_{RF3} applied at the AOM. For the 3-body interaction, the relative phase between the tones is then determined by $\arg(\alpha_1^*\alpha_2) = \phi_{RF3} - \phi_{RF1} = 3 \times \phi_{RF2} - \phi_{RF1}$. An additional phase shift of ϕ_d on ω_{RF2} will then control the relative phase of the 3-body interaction with respect to the Bragg coupling $\arg(\alpha_1^*\alpha_2) - 3\phi_B = 3\phi_d$. The same logic applies to the 4-body interaction with the multiplicative fact 3 replaced by 4 in the above expressions.

3 Momentum state selective readout

To read out the populations in individual momentum states at the end of the experimental sequences, we begin by using microwave π-pulses to map the population back from $|F = 2, m_F = 2\rangle$ to $|F = 1, m_F = 0\rangle$. We then apply a Raman π-pulse (two-photon Rabi frequency 4.2 kHz) for transferring atoms from $|F=1, m_F=0, p_0-\hbar k\rangle$ to $|F=2, m_F=0, p+\hbar k\rangle$. A QND measurement is then used to measure the number of atoms or population as described above. We then blow away the $|F=2\rangle$ atoms with a laser beam resonant with the atomic $|F=2\rightarrow 3'\rangle$ transition. We then measure the population in $p_0+\hbar k$ by applying a Raman π-pulse again with the appropriate two-photon detuning and perform a second QND measurement. The momentum exchange interaction's residual superradiance may transfer atoms into adjacent momentum states $p_0 \pm 3\hbar k$. We iterate the above procedure to measure these populations as well. Thus, on every run of the experiment, we measure the populations in the four momentum states $p_0 \pm \hbar k$ and $p_0 \pm 3\hbar k$.

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